

Physics — Newtonian Mechanics Notes

4.1 Newton's First Law

Newton's First Law, also known as the law of inertia, states that an object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced external force. This law defines the concept of inertia, which is the tendency of an object to resist changes in its state of motion.

Inertial mass is defined as a quantitative measure of an object's resistance to acceleration when a net force is applied. The greater the inertial mass of an object, the smaller the acceleration produced by a given force.

4.2 Newton's Second Law

Newton's Second Law states that the net force acting on an object is equal to the mass of the object multiplied by its acceleration, expressed by the formula $F = ma$. This law quantifies the relationship between force, mass, and acceleration and is one of the most frequently used equations in introductory physics.

When multiple forces act on an object, the net force is the vector sum of all individual forces. A free body diagram is a diagrammatic tool used to visualize all forces acting on an object, which is essential for correctly applying Newton's Second Law in problems involving multiple forces.

4.3 Newton's Third Law

Newton's Third Law states that for every action there is an equal and opposite reaction: whenever object A exerts a force on object B, object B simultaneously exerts a force of equal magnitude and opposite direction on object A. A common misconception is that these action-reaction pairs cancel each other out, but they act on different objects and therefore do not cancel.

4.4 Applications: Friction and Circular Motion

Friction is defined as a force that opposes relative motion or the tendency toward relative motion between two surfaces in contact. Static friction acts on objects that are not moving relative to each other, while kinetic friction acts on objects that are sliding relative to each other. The maximum static friction force is generally greater than the kinetic friction force for the same pair of surfaces.

Uniform circular motion occurs when an object moves in a circular path at a constant speed. Although the speed is constant, the direction of velocity is continuously changing, which means the object experiences a nonzero acceleration called centripetal acceleration, directed toward the center of the circle, given by the formula $a_c = v^2 / r$.

Centripetal force is defined as the net force required to keep an object moving in a circular path, and it

is always directed toward the center of the circle. Students frequently struggle with the concept that centripetal force is not a new, separate type of force but rather the name given to whatever net force is causing the circular motion, such as tension, gravity, or friction.

4.5 Chapter Summary and Review Questions

In summary, Newton's three laws of motion describe the relationship between the forces acting on an object and its resulting motion: the first law describes inertia, the second law quantifies force as the product of mass and acceleration, and the third law describes the pairwise nature of forces between interacting objects.

Review question: Explain why action-reaction force pairs described by Newton's Third Law do not cancel each other out. Review question: How is centripetal force different from a fundamental force like gravity or tension? Review question: What is the formula for centripetal acceleration and what does each variable represent?